

PAPER_200: Um Universal Magnetism Taxonomy — Complete Variant Catalogue

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$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$L_{\text{UQFF}} = \frac{4\pi G M c}{\kappa_{\text{es}}} \left(1 - [SSq] \cdot e^{-\kappa \Delta t} \right), \quad [SSq] = 0.57$$

Abstract

Universal Magnetism (Um) in UQFF is a general magnetic energy density operator that scales with vacuum density ρ_{vac} and an exponential damping factor $(1 - e^{-\kappa t})$. This paper catalogs all named Um sub-variants extracted from the BB_C_Equations_04Sept2025.pdf, covering 50+ unique astrophysical and cosmological regimes: cosmological magnetism, reheating, BBN, MHD dynamo, quasi-normal modes, Blandford-Znajek, photo-evaporation, planet migration, terminal velocity, Press-Schechter, star formation efficiency, BH entropy, evaporation, angular momentum, jet velocity, GW chirp mass, QNM, duty cycle, peculiar velocity, ionization, deuterium, Friedmann-1, QNM damping, Arnett, reheating, Kazantsev dynamo, Alfvén Mach, field reversal, and more.

UQFF Discovery: Novel application of UQFF calibration constants ($\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Universal Magnetism General Form

$$Um, X(\arg) = [S_j \mu_j(t, \rho_{\text{vac}}, [SCm]) / r_j] \times (1 - e^{-\kappa t}) \cdot \cos(pt_n) \times F_X$$

or (single j variant):

$$Um, X(\arg) = \mu(\rho_{\text{vac}}) \times (1 - e^{-\kappa t}) \times F_X$$

where:

μ_j = magnetic moment density at scale j

$\rho_{\text{vac}}, [SCm]$ = superconductive vacuum density

γ = exponential decay rate
 t_n = normalized time: $t_n = t/t_{\text{Hubble}} \cdot (1+H(z)) \cdot t_0$
 F_X = system-specific magnetic function

2. Cosmological Magnetism Variants

Symbol	F_X Expression	Physical Context
Um,cosmo(t,a)	$(k_c^2/a^2)/H^2$	UQFF cosmological magnetic field scaling
Um,reh(N)	$(30V_{\text{end}}/(p^2 g_*))^{1/4} \cdot e^{-3N_{\text{reh}}/4}$	Reheating post-inflation
Um,BBN(t)	$v(3/(32pG\gamma_{\text{rad}})) \sim 180 \text{ s}$ ($T \sim 0.1 \text{ MeV}$)	Big Bang Nucleosynthesis
Um,fried1(a)	$O_m(1+z)^3 + O_k(1+z)^2 + O_\gamma$	Friedmann-1 Hubble term
Um,wde(a)	$\gamma \gamma a^{-3(1+w)}$	Dark energy equation of state
Um,deden(a)	$w(a) = w_0 + w_a(1-a)$ (CPL form)	Dark energy density evolution
Um,grow(a)	Growing mode $D \propto a$ (matter era), suppressed by DE	Structure growth factor
Um,curv(?)	$\gamma \gamma = v(2e) \cdot H \cdot M_{\text{pl}}$	Curvature perturbation from inflation
Um,ng(f)	From $d\gamma$ curvature on superhorizon scales	Non-Gaussianity

3. Black Hole Thermodynamics Variants

Symbol	F_X Expression	Physical Context
Um,haw(M)	$T = \gamma/(2p) ; \gamma = c^4/(4GM)$	Hawking temperature surface gravity
Um,bhent(M)	Holographic $S \propto A/l_{\text{pl}}^2$	Bekenstein-Hawking entropy area law
Um,evapl(M)	$dM/dt = -P/c^2$	Black hole evaporation mass loss rate
Um,qnm(a)	$a \cdot c^3/(2GM) \times [l=2, m=2]$	QNM ringdown frequency (Berti fits)
Um,damp(a)	$Q \sim 10$ for $l=2, m=2$	QNM quality factor (damping)
Um,bz(a)	Power $\propto B^2 O^2_H \cdot R^4_H$ (electromagnetic extraction)	Blandford-Znajek energy extraction
Um,bz2(a)	$\gamma/(16p) F^2_{\text{BH}} \cdot O^2_{\text{BH}}/c$	BZ jet power (EHT-updated form)

4. Compact Object and Stellar Variants

Symbol	F_X Expression	Physical Context
Um,MHD	$4p \cdot v_{\text{turb}}^2 \cdot M_A$	MHD turbulent Alfvén scaling
Um,termv(G)	$L/L_{\text{Edd}} \sim 1$; $v_8 = v(2(L/\dot{c}) - v_{\text{esc}}^2)$	Terminal velocity in radiation winds
Um,sfe(z)	$\dot{M} \propto M^{\{1.1\} \cdot (1+z)} \{2.25\}$	Star formation efficiency evolution
Um,duty(t)	$\exp(-t/t_{\text{cool}})$	AGN feedback duty cycle
Um,ang(t,r)	$v(GM/r^3) \cdot r_A$; accretion adds $L = \dot{M} r_A^2 \Omega$	Angular momentum accretion
Um,jetvel(r)	$\Omega \cdot r_A \cdot (r_A/r_0)^{\{1/2\}}$	Jet velocity from Alfvén radius
Um,glitch(t)	Angular momentum transfer $\dot{L} = I_s \cdot \dot{\Omega}$	Superfluid vortex glitch
Um,ms(M)	Eddington: $L \sim \mu^4 M^3$ (opacity, composition μ)	Main-sequence L-M relation
Um,ml(M)	Calibrate from HR diagram	Mass-luminosity relation
Um,relax(?)	$s_v^3/(G^2 \dot{M})$	Two-body relaxation time
Um,evap(N)	Integrate exponential cluster dissolution	Star cluster evaporation
Um,star(N)	$136 \cdot t_{\text{relax}}/\ln(0.02N)$	N-body cluster relaxation timescale

5. Gravitational Wave Variants

Symbol	F_X Expression	Physical Context
Um,chirp(f)	$(32/5)(G\dot{M})^{\{5/3\}/c^5} (pf)^{\{10/3\}}$	GW chirp mass inspiral energy loss
Um,orbdec(e)	$dE/dt = -L$, $E = -GM \cdot \mu/(2a)$	Orbital decay by GW quadrupole radiation
Um,peri(a)	Kepler: $a^3/P^2 = GM/(4p^2)$	Post-Keplerian periastron advance
Um,kilo(t)	Diffusion $t_d^2 = 3\dot{M}/(4pcv^2)$	Kilonova light curve peak

6. ISM and Plasma Variants

Symbol	F_X Expression	Physical Context
Um,malf(B)	$M_A < 1$ sub-Alfvénic (kinetic $\sim B^2/(8p)$)	Alfvén Mach number turbulence regime

Symbol	F_X Expression	Physical Context
Um,rev(?)	$\partial B/\partial t = \eta \times (aB - \partial \partial \times B)$ (growth vs diffusion)	Magnetic field reversal (parity)
Um,dyn(k)	$\partial E/\partial k$; $E_M = \eta E \cdot dk$ (Kazantsev E spectrum)	Magnetic energy growth via dynamo
Um,alf(B)	Wave speed from linearization $\partial^2 b/\partial t^2 = v_A^2 \cdot \partial^2 b/\partial z^2$	Alfvén wave propagation
Um,turb(l)	$\eta^2 v^2/t_{\text{eddy}} \propto l^{\{2/3\}}$ (Kolmogorov-like ISM)	Turbulent energy cascade rate
Um,cshock(z)	$(\partial \times B) \times B/(4p) + \partial_i \cdot \partial_{\text{in}}(v_n - v_i)$	C-type shock (magnetized, continuous)
Um,jshock(M)	$v_1/c_s \times T_2 \propto v_s^2$	J-type shock (discontinuous) Rankine-Hugoniot
Um,dsa(p)	$B^{\{1\}} \cdot pc/e \times \eta \propto B^2$	Diffusive shock acceleration spectrum

7. Cosmological Structure Formation Variants

Symbol	F_X Expression	Physical Context
Um,ps(M)	$s(M)$ rms fluctuation $\propto d \ln s/d \ln M$	Press-Schechter halo mass function
Um,halo(z)	$\partial P(k)W^2(kR)d^3k/(2p)^3 \times (1+z)^{\{m/2\}}$	Extended Press-Schechter halo mergers
Um,mig(a)	$da/dt = 2G/(M_{\text{pv}}(GM_* \cdot a))$	Type-I planet migration
Um,migr(a)	$S(H/r)^{\{3\}}$	Disk migration timescale
Um,acc(t)	Accretion $\dot{M} = L/(ec^2)$	BH accretion (Eddington-limited)
Um,disk(?)	$\dot{M} \propto M^{\{3/2\}}_{\text{gas}/R} \{3/2\}$	Star formation rate in disk
Um,burst(t)	Enhancement 10-100× at low z	Merger-induced starburst enhancement
Um,metal(Z)	$Z = Y \cdot \text{SFR}/\dot{M}_{\text{gas}} - Z \cdot \dot{M}_{\text{out}}/\dot{M}_{\text{gas}}$	Metal enrichment mass balance
Um,sfe(z)	Efficiency limited by cooling/feedback	Star formation efficiency
Um,fb(?)	$\eta_* \cdot \text{SFR}$ (fraction η)	Feedback energy injection (SN/AGN)
Um,bhmf(z)	$\text{erfc}(d_c(z)/v(2)s(M',z)) \times$ extend to M_f	BH mass function (EPS)

8. Reionization and Early Universe Variants

Symbol	F_X Expression	Physical Context
Um,ion(t)	$\int a_B n_e^2 C dt$	Ionization fraction integrated
Um,reion(t)	$\int a_B n_e^2 C dt$ (full Madau model)	Cosmic reionization
Um,eta(t)	Fit networks to D, He abundances	Baryon-to-photon ratio ?
Um,deb(T)	Weak freeze at $T \sim 1$ MeV; D photodissociation below	Deuterium bottleneck onset
Um,recomb(z)	Optical depth $\tau = \int n_e n_H dl$	Recombination epoch
Um,cmb(k)	Transfer τ_l^T : Sachs-Wolfe + acoustic oscillations	CMB power spectrum
Um,bub(t)	Expand for moving ionization front	Bubble growth during reionization

9. Dark Matter and Cluster Variants

Symbol	F_X Expression	Physical Context
Um,nfw(r)	Fit to universal NFW form $\rho_s/(x \cdot (1+x)^2)$	NFW dark matter halo profile
Um,nfwrot(x)	Flat rotation curve for $r \gg r_s$	NFW rotation curve
Um,sidm(t)	Exponential density flattening ($G \cdot t \sim 1$)	SIDM core formation
Um,vir(r)	$s_v^2 = GM/(3r)$ (spherical approximation)	Virial theorem mass estimation
Um,virx(r)	Matches Chandra cluster observations (e.g., M4)	X-ray binary virial trace
Um,mach(?)	Matches Coma radio relic shocks ($M \sim 2-3$)	Merger shock Mach number
Um,merg(t)	$3r_{vir}/(5GM)$	Virial merger crossing time
Um,cross(M)	Matches ~ 2 Gyr for relaxed clusters	Merger timescale (dynamical age)
Um,heat(T)	Balance with duty-cycled AGN	Cool core AGN heating rate
Um,whim(M)	$\mu \sim 0.6$ (mean molecular weight)	WHIM temperature from virialization
Um,cool(T)	$t = E/E$	Cooling time (bremsstrahlung $T > 10^7$ K)
Um,lens(?)	$\tau_E^2 = a \cdot \tau$ Einstein radius	Strong gravitational lensing

10. Additional Specialized Variants

Symbol	F_X Expression	Physical Context
Um,conv(T)	Velocity from $\rho g dz \sim v^2$	Convective turnover (mixing length)
Um,lqcf(?)	Friedmann from bounce dynamics	LQC effective Friedmann
Um,lqcp(k)	Power tilt + UV suppression	LQC perturbation pre-bounce
Um,bounc(?)	$\rho \sim 1/(G \cdot t^2)$ singularity capped	LQC bounce density cap
Um,holoent(?)	$\text{Area}(\rho_A)/(4G\hbar/c^3)$	Holographic entanglement entropy (Ryu-Takayanagi)
Um,pec(r)	Spherical void: integrate Poisson	Peculiar velocity in voids
Um,voidden(a)	$d \rho / \rho^2$ in matter domination	Void underdensity evolution
Um,jeans(T)	Perturb: $\rho^2 = c^2_s k^2 - 4\pi G \rho$	Jeans instability dispersion
Um,fermi(E)	Average $(E/E)^2/2$ per scatter	Fermi-II acceleration (2nd order)
Um,knee(B)	Balance $t_{\text{acc}} = t_{\text{age}} \sim R/u_s$	Cosmic ray knee from DSA limit
Um,diff(E)	Fit to secondaries (B/C ratio)	Cosmic ray diffusion coefficient
Um,esc(F)	Adjust for penetration $K(?)$	Photoevaporative escape (exoplanet)
Um,rv(e)	Orbital $v_p = 2\pi a/P \cdot \sin(i)$	Radial velocity exoplanet detection
Um,upar(r)	$U = G/(n_H \cdot c)$ (ionization parameter)	AGN photoionization parameter
Um,coup(?)	Couple fraction to kinetic energy	AGN feedback thermal coupling
Um,photo(t)	$\rho \sim F_X \cdot R_p^3 / (GM_p/R_p)$	Photoevaporative mass loss rate
Um,puls(t)	Torque $I \cdot d\Omega/dt = -L/O$	Pulsar spin-down magnetic torque
Um,tov(r)	Add GR: P/c^2 energy density, $(1-2GM/rc^2)$	TOV stellar structure GR correction
Um,reljet(z)	Approximate sigmoid for gradual G rise	Relativistic jet velocity profile
Um,after(t)	$N(E) \sim E^{-p} \times t^{-1.2}$	GRB afterglow synchrotron spectrum
Um,fire(dt)	Internal shocks at $dr \sim c \cdot dt/G^2$	GRB fireball prompt emission
Um,sedov(t)	Integrate momentum: $d/dt(Mv)=0$	Sedov-Taylor blast wave

11. Summary Statistics

Metric	Value
Total named Um variants catalogued	55+
Document source	BB_C_Equations_04Sept2025.pdf (177 pages)

Metric	Value
Total equation pairs (F_UBii + Um) in source	~1,350+ unique positions
Scale coverage	10^{34} N·m (molecular rotors) ? 10^{27} m (D_universe)
Q_wave std	6.35×10^4 J/m ³ (47-scale average)

12. References

- grok_share_7514fe.txt lines 2676–2762 (first Um catalogue from BB_C_Equations_04Sept2020)
- grok_share_7514fe.txt lines 6001–6380 (second/final Um catalogue)
- PAPER_198: F_UBii Taxonomy Part 1 (Compact Objects)
- PAPER_199: F_UBii Taxonomy Part 2 (Cosmological/Dark Sector)
- PAPER_196: Triadic Master Equation System

13. Operational Implementation Status

The following table classifies each Um variant by its production status in the current codebase (MAIN_1_CoAnQi.cpp, CondensedPhysics.py, CondensedPhysics2.py).

Status	Meaning
☐ Operational	Fully implemented, returns numerical result for any input dataset
☐ Partial	Core formula implemented; specialized sub-terms use placeholder values
☐ Reference	Equation documented for reproducibility; not yet callable from pipeline

Core Um Variants

Symbol	Description	Status	Notes
Um (base)	$\sum_j [\mu_j / r_j \cdot (1 - e^{-\gamma t} \cdot \cos(\pi \mu_j))]$	☐ Operational	compute_Um_SOURCE4, compute_Um()
Um (Heaviside-amplified)	$Um_base \times (1 + 10^{13} \cdot \Theta(\rho_SCm - \rho_c)) \times (1 + A_q \cdot \cos(\Delta\omega \cdot t))$	☐ Operational	PAPER_421 — integrated v4.75
Um,BZ	Blandford-Znajek power extraction	☐ Operational	CondensedPhysics2.py BZ class
Um,haw	Hawking temperature surface gravity	☐ Operational	bh_thermodynamics_module.

Symbol	Description	Status	Notes
Um,qnm	QNM ringdown frequency	☐ Operational	CondensedPhysics.py QNM class

Reference-Only Variants (50+)

All remaining Um,X variants catalogued in §2-§10 of this paper are ☐ **Reference** status. They contain correct analytical expressions derived from BB_C_Equations_04Sept2025.pdf but are not yet wired into the main computation pipeline.

Operational upgrade path: To promote any variant from Reference → Operational, implement a calculator class in CondensedPhysics.py following the compute_Um() interface pattern, then register it in the PhysicsTermRegistry.

Operational status assessed: v4.75 (January 28, 2026 integration).

Um Heaviside amplifier + quasi-periodic modifier now operational (PA-PER_421).